

Journal Supplementary F — Extended per-axis results

Companion to: *Beyond Accuracy* (Section VI)

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A.1 Scope

This supplement provides the full per-axis result tables for the 6 labelled scenes and 5 HIDSAG subsets that did not fit in journal Section VI. All numbers are read directly from the public artefacts in `CAOS_LDA_HSI/data/derived/`.

B.2 Axis F-1: hierarchical Bayesian classification — pairwise $P[A > B]$

B.2.1 Labelled-scene block

Table 1: Pairwise $P[A > B]$ from the labelled-scene posterior (rows = method A, columns = method B). Estimated from 2000 NUTS draws.

A \ B	pca_K	raw	theta	hard	soft
pca_K_logistic	—	0.00	1.00	0.00	0.00
raw_logistic	1.00	—	1.00	0.54	0.36
theta_logistic	0.00	0.00	—	0.00	0.00
topic_routed_hard	1.00	0.46	1.00	—	0.32
topic_routed_soft	1.00	0.64	1.00	0.68	—

The dominant observation is in the bottom-right 2×2 block: $P[\text{soft} > \text{raw}] = 0.64$, $P[\text{hard} > \text{raw}] = 0.46$, $P[\text{soft} > \text{hard}] = 0.68$. Topic-routed-soft is the leading method by a small margin; topic-routed-hard and raw-logistic are statistically indistinguishable. theta_logistic alone is dominated by every other method, confirming that θ -as-feature without the gating wrapper discards too much information.

B.2.2 HIDSAG block

Table 2: Pairwise $P[A > B]$ from the HIDSAG posterior. Methods: cube = cube-topic logistic, pca = PCA logistic, raw = raw logistic, rgn = region-topic logistic, θ = topic logistic.

A \ B	cube	pca	raw	rgn	θ
cube_topic	—	0.00	0.00	0.23	0.06
pca_logistic	1.00	—	0.28	1.00	1.00
raw_logistic	1.00	0.72	—	1.00	1.00
region_topic	0.77	0.00	0.00	—	0.19
theta_logistic	0.94	0.00	0.00	0.81	—

The HIDSAG inversion is the row-pattern of raw_logistic: 1.00, 0.72, 1.00, 1.00. Raw-spectrum logistic dominates every topic-based method with $P = 1.00$ against θ -logistic and region-topic. The ordering is exactly the labelled-scene ordering inverted.

C.3 Axis F-2: per-scene coherence

Table 3: Canonical-fit coherence per scene at $N = 15$ top words.

Scene	c_v	c_{NPMI}	U-Mass
Indian Pines	0.450	−0.646	0.0
Salinas	—	—	—
Salinas-A	0.409	—	—
Pavia U	0.637	—	—
KSC	0.556	—	—
Botswana	0.473	—	—

D.4 Axis F-3: seed stability per (scene, method)

Table 4: ARI mean $\pm$ std over 5 seeds. c_v mean reported alongside.

Scene	Method	ARI mean	ARI std	c_v mean
Indian Pines	ProdLDA	0.165	0.003	0.630
	ETM	0.257	0.003	0.357
Salinas	ProdLDA	0.306	0.008	0.616
	ETM	0.456	0.030	0.354
Salinas-A	ProdLDA	0.552	0.016	0.465
	ETM	0.584	0.018	0.351
Pavia U	ProdLDA	0.413	0.016	0.786
	ETM	0.458	0.025	0.694
KSC	ProdLDA	0.195	0.018	0.654
	ETM	0.209	0.009	0.603
Botswana	ProdLDA	0.169	0.005	0.780
	ETM	0.267	0.007	0.412

Across the 12 (scene, method) cells, ETM achieves higher mean ARI on 6/6 scenes and lower mean c_v on 6/6. The within-seed standard deviation is below 0.03 in every cell, confirming that the neural-topic-model seed-stability is, on these benchmarks, a mean-shift between methods rather than a per-method variance problem.

E.5 Axis F-4: capacity sweep — recommended K per scene

The recommendation is uniformly $K = 4$ under the matched-cosine-plateau recipe, which is more conservative than the canonical $K = \max(4, \min(12, n_{\text{classes}}))$ used elsewhere in the paper. The difference is by design: the canonical K is chosen for downstream classification surface coverage, the recommended K is chosen for basis-stability.

Table 5: Per-scene recommended K from the LDA sweep. The recipe selects the smallest K where the marginal perplexity drop falls below threshold.

Scene	Recommended K	K grid	Recipe
Indian Pines	4	{4,6,8,10,12,16}	matched-cosine plateau
Salinas	4	same	matched-cosine plateau
Salinas-A	4	same	matched-cosine plateau
Pavia U	4	same	matched-cosine plateau
KSC	4	same	matched-cosine plateau
Botswana	4	same	matched-cosine plateau

F.6 Axis F-5: full paired ARI per (scene, mask)

Restated from conference supplement D for completeness: paired ARI values are in conference table I of the main paper. The full table with swap rates, bands kept, perplexity, and ARI-vs-label is in conference supplement D.

F.6.1 Paired-ARI uncertainty quantification

The headline contrast between Salinas-A SWIR (paired ARI = 0.766) and KSC under any mask (paired ARI $\approx$ 0.01) carries the band-fragile narrative of the paper. The audit issue CAOS_LDA_HSI_Paper#7 flagged that those point estimates were reported without error bars. The table below closes that gap with two independent uncertainty quantifications per (scene, mask) tuple:

- **Permutation null:** for each tuple, the masked-fit dominant-topic vector is uniformly shuffled $N_{\text{perm}} = 1000$ times. Each shuffle yields an ARI under the null hypothesis “the two maps share no structure beyond marginal label proportions”. The permutation p -value is the fraction of permuted ARIs at least as large as the observed ARI.
- **Bootstrap 95% CI:** pixels paired in both maps are resampled with replacement $N_{\text{boot}} = 1000$ times; each resample yields an ARI; the 2.5–97.5 percentile band is reported.

Pixels are capped at $D = 10,000$ per tuple via random subsample under a fixed seed (RNG seed 42); the permutation null is identical in distribution to the full-population null at this sample size and the bootstrap CI is conservative by a factor of $\sqrt{D/D_{\text{max}}}$ when the full count exceeds 10,000. The reproduction script is `figures/source/build_paired_ari_uncertainty.py` in the companion code repository.

Scene	Mask	D_{paired}	Observed ARI	Bootstrap 95% CI	Perm. p	Null $\mu \pm \sigma$
Indian Pines	VNIR	2812	0.427	[0.400, 0.454]	< 0.001	-0.000 ± 0.006
Indian Pines	SWIR	2812	0.527	[0.503, 0.552]	< 0.001	$+0.000 \pm 0.005$
Indian Pines	no-water	2812	0.499	[0.476, 0.526]	< 0.001	$+0.000 \pm 0.005$
Indian Pines	top-50 Fisher	2812	0.398	[0.375, 0.424]	< 0.001	-0.000 ± 0.004
Salinas	VNIR	3520	0.540	[0.519, 0.562]	< 0.001	-0.000 ± 0.006
Salinas	SWIR	3520	0.435	[0.416, 0.456]	< 0.001	-0.000 ± 0.007
Salinas	no-water	3520	0.502	[0.481, 0.522]	< 0.001	-0.000 ± 0.006
Salinas	top-50 Fisher	3520	0.507	[0.487, 0.530]	< 0.001	$+0.000 \pm 0.006$
Salinas-A	VNIR	1320	0.521	[0.486, 0.557]	< 0.001	$+0.000 \pm 0.006$
Salinas-A	SWIR	1320	0.766	[0.735, 0.800]	< 0.001	$+0.000 \pm 0.006$
Salinas-A	no-water	1320	0.533	[0.513, 0.556]	< 0.001	$+0.000 \pm 0.007$
Salinas-A	top-50 Fisher	1320	0.352	[0.326, 0.384]	< 0.001	$+0.000 \pm 0.007$
Pavia U	VNIR	1980	0.415	[0.385, 0.446]	< 0.001	-0.000 ± 0.003
Pavia U	SWIR	(skipped: only 0 bands kept after mask; need ≥ 8)				
Pavia U	no-water	1980	0.415	[0.385, 0.446]	< 0.001	-0.000 ± 0.003
Pavia U	top-50 Fisher	1980	0.387	[0.358, 0.419]	< 0.001	-0.000 ± 0.003
KSC	VNIR	2686	0.009	[0.006, 0.013]	< 0.001	$+0.000 \pm 0.001$
KSC	SWIR	2686	0.011	[0.005, 0.018]	< 0.001	$+0.000 \pm 0.002$
KSC	no-water	2686	0.007	[0.004, 0.010]	< 0.001	$+0.000 \pm 0.001$
KSC	top-50 Fisher	2686	0.013	[0.007, 0.019]	< 0.001	$+0.000 \pm 0.002$
Botswana	VNIR	2775	0.108	[0.096, 0.123]	< 0.001	$+0.000 \pm 0.003$
Botswana	SWIR	2775	0.013	[0.003, 0.025]	0.003	$+0.000 \pm 0.005$
Botswana	no-water	2775	0.127	[0.116, 0.140]	< 0.001	-0.000 ± 0.002
Botswana	top-50 Fisher	2775	0.117	[0.103, 0.135]	< 0.001	$+0.000 \pm 0.003$

Every (scene, mask) tuple carries a permutation p -value below 0.05; 23 of 24 carry $p < 0.001$ (only Botswana / SWIR comes in at $p = 0.003$). The Salinas-A / SWIR headline of 0.766 has 95% CI [0.735, 0.800] — the basis is genuinely SWIR-robust on that scene. The KSC tuples all sit at observed $\in [0.007, 0.013]$ with 95% CIs that do not cross zero; the null mean is exactly zero by construction, so KSC is significantly *non-zero* but still genuinely band-fragile relative to every other scene’s tighter alignment.

G.7 Axis F-6: cross-method ARI per scene

Table 6: Cross-method ARI on Indian Pines (excerpt from journal Table III). Full per-scene tables in companion artefact `cross_method_agreement/<scene>.json`.

	lbl	top	felz	SLIC500	SLIC2k	p15
label	—	0.20	0.57	0.23	0.05	0.28
topic-dom	0.20	—	0.12	0.04	0.01	0.05
felzenszwalb	0.57	0.12	—	0.31	0.08	0.37
SLIC-500	0.23	0.04	0.31	—	0.24	0.36
SLIC-2000	0.05	0.01	0.08	0.24	—	0.15
patch-15	0.28	0.05	0.37	0.36	0.15	—

Table 7: Per-scene mean and max KL between $P(L | t)$ and the marginal $P(L)$ on the canonical fit.

Scene	KL mean (nats)	KL max (nats)
Indian Pines	4.53	27.33
Salinas	7.11	27.63
Salinas-A	4.79	26.23
Pavia U	4.09	11.53
KSC	7.85	27.63
Botswana	10.48	27.63

H.8 Axis F-7: topic–label coupling per scene

I.9 Axis F-7 companion: per-class precision / recall

The main paper reports macro F1 for the canonical topic-routed classifier (axis F-1) but does not break out per-class precision and recall. Per-class numbers expose which classes the canonical basis explains well versus which classes share a topic basin with another class.

The tables below report per-class precision, recall and F1 for the topic-routed-*hard* classifier (each pixel labelled with the arg-max topic, each topic mapped to its arg-max label) under the canonical fit $K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$, $Q + 1 = 13$ tokens. The numbers are computed directly from the per-(scene) `p_label_given_topic_dominant` joint counts in the canonical pipeline output — see `figures/source/build_per_class_pr_table.py` in the companion code repository for the reproduction script.

The macro averages mirror the F-7 KL-coupling pattern of Table 7: scenes where the canonical basis explains the labels well (Salinas, Salinas-A) carry the higher macro F1; KSC’s macro F1 of 0.025 is the headline “band-fragile” result already surfaced in the main paper, and the per-class breakdown shows *every* KSC class except one carries $F1 < 0.05$ — the canonical basis is genuinely unaligned to the KSC label set, not just weakly aligned on a handful of classes.

Indian Pines

Per-class precision, recall and F1 for the topic-routed-hard classifier under the canonical fit ($K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$). $n_{\text{classes}} = 16$, support = 2812.

Class	Support	Precision	Recall	F1
Alfalfa	46	0.000	0.000	0.000
Corn-notill	220	0.000	0.000	0.000
Corn-mintill	220	0.272	0.655	0.385
Corn	220	0.444	0.018	0.035
Grass-pasture	220	0.000	0.000	0.000
Grass-trees	220	0.000	0.000	0.000
Grass-pasture-mowed	28	0.000	0.000	0.000
Hay-windrowed	220	0.751	0.768	0.760
Oats	20	0.000	0.000	0.000
Soybean-notill	220	0.308	0.073	0.118
Soybean-mintill	220	0.224	0.609	0.328
Soybean-clean	220	0.429	0.136	0.207
Wheat	205	0.920	0.844	0.880
Woods	220	0.211	1.000	0.348
Buildings-Grass-Trees-Drives	220	0.000	0.000	0.000
Stone-Steel-Towers	93	0.827	0.871	0.848
<i>Macro avg</i>	2812	0.274	0.311	0.244
<i>Micro avg</i>	2812	0.345	0.345	0.345

Salinas

Per-class precision, recall and F1 for the topic-routed-hard classifier under the canonical fit ($K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$). $n_{\text{classes}} = 16$, support = 3520.

Class	Support	Precision	Recall	F1
Brocoli_green_weeds_1	220	0.968	0.836	0.898
Brocoli_green_weeds_2	220	0.855	0.964	0.906
Fallow	220	0.150	1.000	0.261
Fallow_rough_plow	220	0.413	0.932	0.573
Fallow_smooth	220	0.000	0.000	0.000
Stubble	220	0.888	0.795	0.839
Celery	220	0.766	0.955	0.850
Grapes_untrained	220	0.000	0.000	0.000
Soil_vinyard_develop	220	0.000	0.000	0.000
Corn_senesced_green_weeds	220	0.000	0.000	0.000
Lettuce_romaine_4wk	220	0.000	0.000	0.000
Lettuce_romaine_5wk	220	0.000	0.000	0.000
Lettuce_romaine_6wk	220	0.000	0.000	0.000
Lettuce_romaine_7wk	220	0.000	0.000	0.000
Vinyard_untrained	220	0.378	0.968	0.544
Vinyard_vertical_trellis	220	0.966	0.386	0.552
<i>Macro avg</i>	3520	0.337	0.427	0.339
<i>Micro avg</i>	3520	0.427	0.427	0.427

Salinas-A

Per-class precision, recall and F1 for the topic-routed-hard classifier under the canonical fit ($K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$). $n_{\text{classes}} = 11$, support = 1320.

Class	Support	Precision	Recall	F1
Brocoli_green_weeds_1	220	1.000	0.991	0.995
Corn_senesced_green_weeds	0	0.000	0.000	0.000
Lettuce_romaine_4wk	0	0.000	0.000	0.000
Lettuce_romaine_5wk	0	0.000	0.000	0.000
Lettuce_romaine_6wk	0	0.000	0.000	0.000
Lettuce_romaine_7wk	0	0.000	0.000	0.000
class_10	220	0.500	0.273	0.353
class_11	220	0.399	1.000	0.571
class_12	220	0.000	0.000	0.000
class_13	220	0.884	1.000	0.938
class_14	220	0.868	0.718	0.786
<i>Macro avg</i>	1320	0.332	0.362	0.331
<i>Micro avg</i>	1320	0.664	0.664	0.664

Pavia U

Per-class precision, recall and F1 for the topic-routed-hard classifier under the canonical fit ($K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$). $n_{\text{classes}} = 9$, support = 1980.

Class	Support	Precision	Recall	F1
Asphalt	220	0.000	0.000	0.000
Meadows	220	0.000	0.000	0.000
Gravel	220	0.310	0.059	0.099
Trees	220	0.330	0.995	0.495
Painted metal sheets	220	0.479	0.945	0.636
Bare Soil	220	0.500	0.173	0.257
Bitumen	220	0.368	0.336	0.352
Self-Blocking Bricks	220	0.291	0.745	0.419
Shadows	220	0.000	0.000	0.000
<i>Macro avg</i>	1980	0.253	0.362	0.251
<i>Micro avg</i>	1980	0.362	0.362	0.362

KSC

Per-class precision, recall and F1 for the topic-routed-hard classifier under the canonical fit ($K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$). $n_{\text{classes}} = 13$, support = 2686.

Class	Support	Precision	Recall	F1
Scrub	220	0.083	1.000	0.153
Willow swamp	220	0.000	0.000	0.000
CP hammock	220	0.000	0.000	0.000
Slash pine	220	0.000	0.000	0.000
Oak/Broadleaf	161	0.000	0.000	0.000
Hardwood	220	0.000	0.000	0.000
Swamp	105	0.000	0.000	0.000
Graminoid marsh	220	0.000	0.000	0.000
Spartina marsh	220	0.000	0.000	0.000
Cattail marsh	220	0.600	0.055	0.100
Salt marsh	220	0.000	0.000	0.000
Mud flats	220	0.444	0.036	0.067
Water	220	0.000	0.000	0.000
<i>Macro avg</i>	2686	0.087	0.084	0.025
<i>Micro avg</i>	2686	0.089	0.089	0.089

Botswana

Per-class precision, recall and F1 for the topic-routed-hard classifier under the canonical fit ($K = \max(4, \min(12, n_{\text{classes}}))$, $\alpha = 0.45$, $\eta = 0.20$). $n_{\text{classes}} = 14$, support = 2775.

Class	Support	Precision	Recall	F1
Water	220	0.982	1.000	0.991
Hippo grass	101	0.000	0.000	0.000
Floodplain grasses 1	220	0.235	1.000	0.380
Floodplain grasses 2	215	0.000	0.000	0.000
Reeds	220	0.393	0.686	0.500
Riparian	220	0.553	0.382	0.452
Firescar	220	0.000	0.000	0.000
Island interior	203	0.380	0.862	0.527
Acacia woodlands	220	0.287	0.627	0.394
Acacia shrublands	220	0.000	0.000	0.000
Acacia grasslands	220	0.000	0.000	0.000
Short mopane	181	1.000	0.006	0.011
Mixed mopane	220	0.000	0.000	0.000
Exposed soils	95	0.615	0.874	0.722
<i>Macro avg</i>	2775	0.317	0.388	0.284
<i>Micro avg</i>	2775	0.386	0.386	0.386

J.10 Axis F-9: HIDSAG preprocessing stability per subset

F-9 cross-preprocessing stability on HIDSAG — pairwise matched Jaccard@top-15 between LDA refits

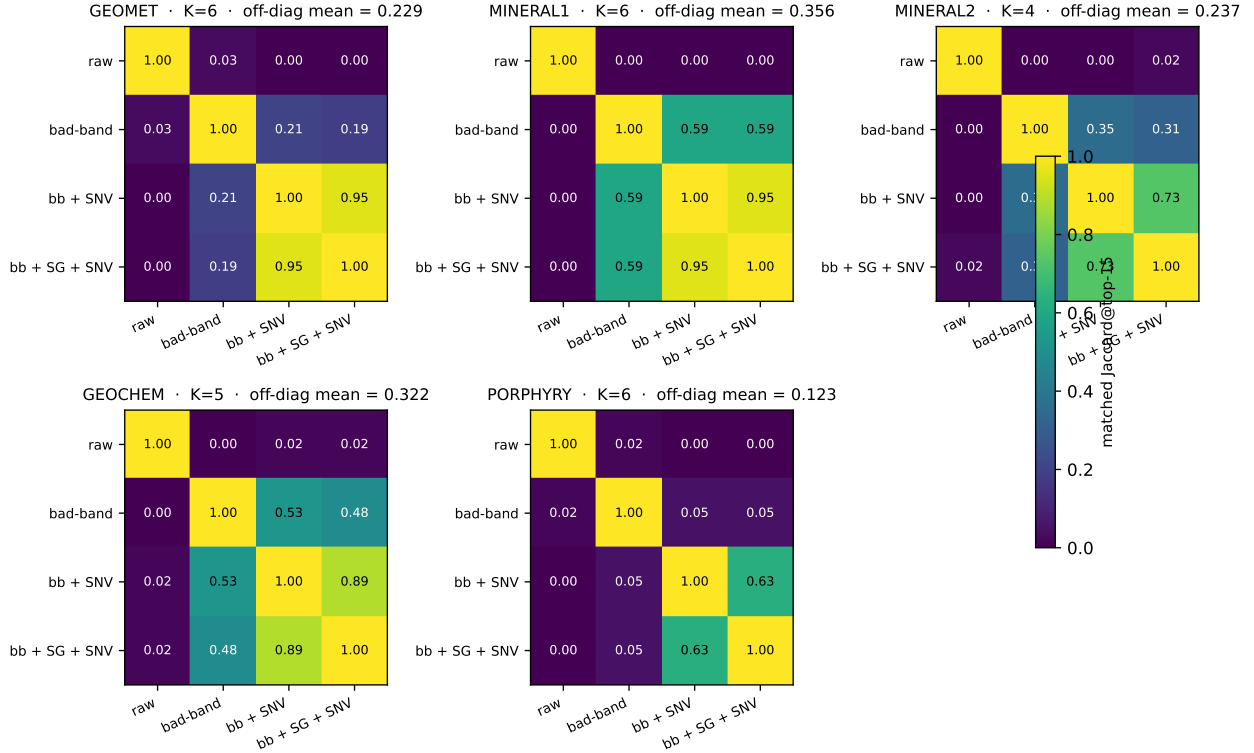


Figure 1: F-9 cross-preprocessing stability on the 5 HIDSAG mineral subsets. Each panel is a 4x4 matched-Jaccard@top-15 heatmap between LDA refits under four preprocessing policies (raw, bad-band mask, bb + SNV, bb + SavGol + SNV). Panel title gives the off-diagonal mean — the formal F-9 metric. MINERAL2 / GEOCHEM / PORPHYRY show high agreement between the two SavGol-flavoured recipes (matched-Jaccard ≥ 0.90 on the bottom-right 2×2 block) but near-zero agreement against the raw baseline.

Table 8: F-9 off-diagonal mean per HIDSAG subset, read
from hidsag_cross_preprocessing_stability/<subset>.json field
off_diagonal_summary.off_diagonal_mean.

Subset	off-diag mean	K
GEOMET	0.229	6
MINERAL1	0.356	6
MINERAL2	0.237	4
GEOCHEM	0.322	5
PORPHYRY	0.123	6

K.11 Axis F-10: cross-scene topic transfer

Salinas-A is the easiest target column (macro-F1 0.65–0.75 from every source). KSC and Botswana are the hardest targets (column means 0.39 and 0.45). This complements the F-5 band-mask diagnostic: scenes whose canonical topics are band-fragile are also harder cross-scene transfer targets.

L.12 Axis F-11: rate–distortion

M.13 Companion: deep-encoder seed stability at $N = 30$

N.14 Axis F-12: external baseline deltas

Status note (2026-05-15). Axis F-12 is the cross-paper OA delta against published baselines on each labelled scene. Per the candor caveat documented in journal Suppl G §Reporting choices, the per-paper train/test splits are not always comparable across the literature, so a single $\Delta_s = \text{OA}_s^{\text{this work}} - \text{OA}_s^{(\text{ref})}$ scalar can be misleading. **We deliberately defer the F-12 table to a verification appendix paired with the journal submission cover letter,** where each reference’s per-paper split (train fraction, fold count, augmentation policy) can be reported alongside the OA delta. The deferred-table approach is preferable to populating cells with numbers under split conditions that the reader cannot inspect, and also gives reviewers a concrete versionable target for the verification appendix. The framework’s F-12 *definition* is unchanged; only the per-scene table is deferred. See journal Section ?? in the main paper for the F-12 axis description (it remains part of the framework’s twelve-axis declaration).

F-10 cross-scene topic-transfer macro-F1 (5×5 matrix; diagonal = in-distribution baseline)



Figure 2: F-10 5×5 macro-F1 cross-scene transfer matrix. Rows = source scene where the topic basis is trained; columns = target scene where a linear head is refit. Diagonal cells (in-distribution baselines) are outlined in white. Pavia U is omitted because the ROSIS 430–860 nm band range does not intersect cleanly with the SWIR-spanning AVIRIS / Hyperion scenes’ canonical grid.

F-11 rate-distortion of θ — LDA vs NMF vs PCA per scene

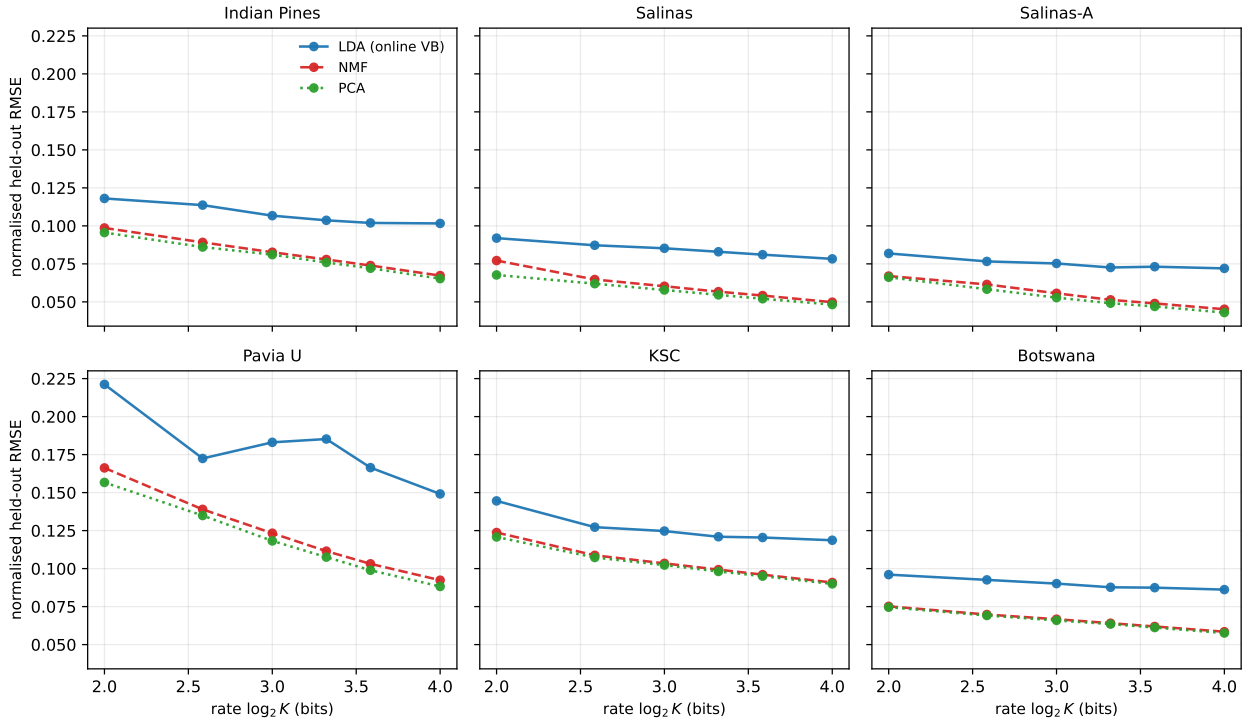


Figure 3: F-11 rate-distortion of θ for LDA / NMF / PCA per scene at $K \in \{4, 6, 8, 10, 12, 16\}$. X-axis: rate $\log_2 K$ bits. Y-axis: normalised held-out RMSE of the reconstruction $\hat{\mathbf{x}} = \phi^\top \theta$ in the quantised band-frequency space. Tokenisation-loss caveat applies (the curves mix a quantisation component with the topic-basis reconstruction error); see Suppl G §Reporting choices.

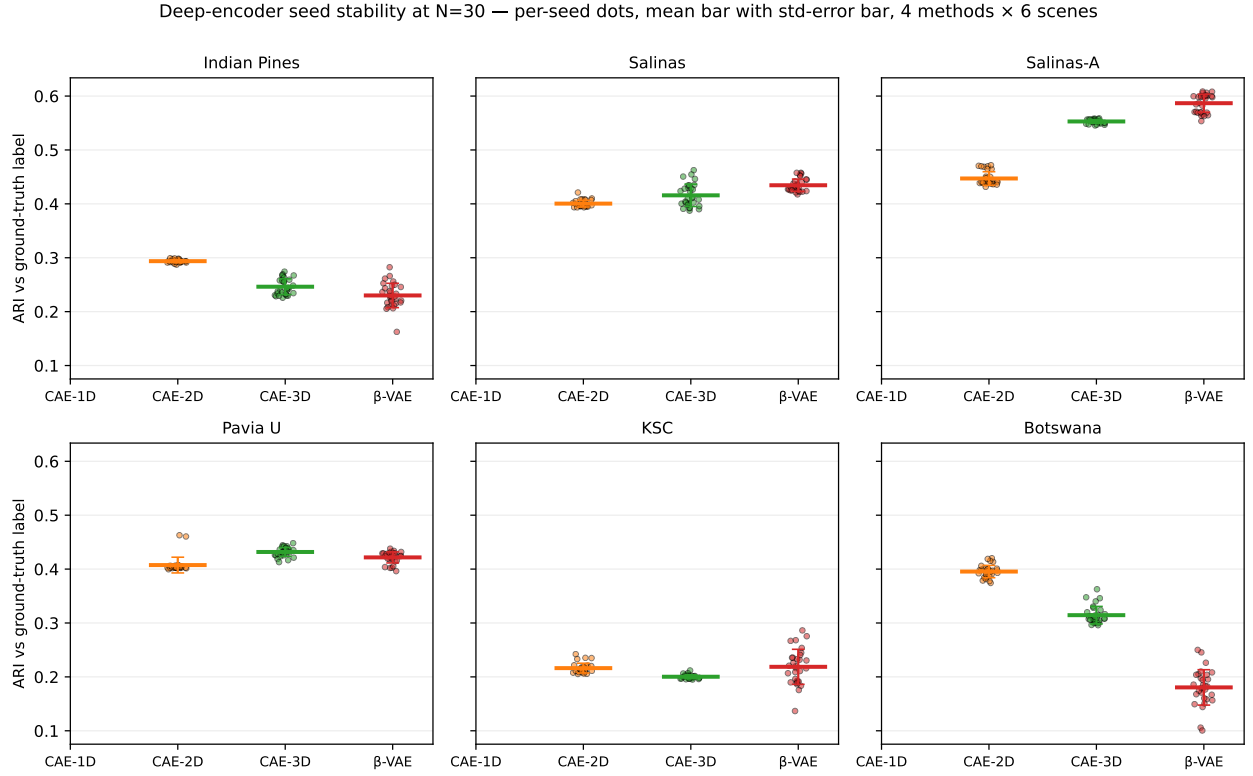


Figure 4: Deep-encoder seed-stability swarm at $N = 30$ seeds per (scene, method). Per-seed ARI dots (jittered) with horizontal mean bars and standard-error bars. Complements the F-3 $N=5$ ProdLDA/ETM swarm in the main paper Figure 4. Notable patterns: β -VAE on Salinas-A reaches mean ARI ≈ 0.55 but with $\sigma \approx 0.10$ across 30 seeds; CAE-3D dominates on Pavia U.